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Opinion paper

# What is novel about the Metaverse?

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ABSTRACT

The Metaverse is not a radical departure. It is an incremental evolution. Previous virtual worlds, like Second Life, have laid the groundwork for the development of the Metaverse by providing valuable insights into virtual environments and social interactions. Studies of these earlier platforms contribute to a more nuanced understanding of what the Metaverse is, and what it is not. Based on studies of virtual worlds and by applying inductive processes of reflection and abstraction, we propose a framework that supports the identification of attributes that help differentiate the Metaverse from previous virtual worlds from a user experience perspective. We demonstrate the value of the framework by comparing the Metaverse to Second Life. Our framework and comparative analysis ground the current discussions about the Metaverse deeper in the established discourse on virtual worlds. They also shed light on the potential impact of the Metaverse, the special role of its immersiveness and help us to identify lessons learned that can inform the Metaverse's further development.

## 1. Introduction

The concept of the Metaverse, which is still evolving, has captured the imagination of technologists, researchers, and the general public alike, with its potential to revolutionize the way we interact and engage with each other in virtual environments (Ball, 2022; Cheng, Wu, Varvello, Chen, & Han, 2022; Dwivedi et al., 2022). Just as some analysts estimate that the global Metaverse market could reach $800 billion by 2024 (IntelligenceBloomberg, 2022), others see it doomed like other virtual worlds before. While the idea holds promise, it is crucial to emphasize that a fully functioning Metaverse has not yet come into existence (Nickerson, Seidel, Yepes, & Berente, 2022). In fact, there are varying interpretations of what the Metaverse is, as can be seen in the strategies of two major players in the field: Facebook's holding Meta Platforms, Inc., focusses on the development on virtual reality (VR) glasses (e.g. Meta Quest Pro) and an own platform (Horizon Workrooms) as part of the Metaverse. Apple's current focus is on augmented reality (AR), i.e. building a virtual layer on top of the real world, rather than recreating the real world virtually. Consequently, Apple has recently released the Vision Pro glasses which are heralded by some as redefining the metaverse, merging immersive digital experiences with the real world through next-level augmented reality (Sorkin et al., 2023).

Discussions and expectations surrounding the Metaverse should be tempered with the understanding that the realization of this groundbreaking digital disruption remains a work in progress, with many challenges and hurdles yet to be addressed in its development and implementation (Mogaji, Wirtz, Belk, & Dwivedi, 2023). Hence, despite the excitement and optimism surrounding the Metaverse, there is a significant gap in our understanding of its attributes and mechanisms.

The enthusiasm surrounding the Metaverse is reminiscent of the hype around Second Life (SL) (Henninger, Bürklin, & Parker, 2019; Kanamgotov, Christopoulos, Conrad, & Prakoonwit, 2012). This virtual world was popular in the early 2000 s and is now often cited as an early example of an early metaverse (Kaplan & Haenlein, 2009). At its peak, SL had over a million active users (Sparkes, 2021), was the subject of much media attention and was seen as a revolutionary new platform for social interaction and commerce (Cagnina & Poian, 2007). However, the publicity surrounding SL eventually died down as the platform struggled to retain users and generate revenue. The substantial cost of operating virtual worlds, combined with the expense of maintaining meeting spaces made it difficult for third-party entities to achieve any significant return on investment (ROI) (Henninger et al., 2019). As a result, it is now a fraction of its former size. After all the attention (Dolata & Schwabe, 2023), will that also be the Metaverse's fate? Or is the Metaverse decisively different from previous versions of virtual worlds such as SL?

With our study, we suggest and apply a framework that contributes to a better conceptual understanding of virtual worlds and their attributes. Hence, the framework helps to understand why users come back to or use virtual worlds in the first place. When validating the

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framework, we draw on a comparative analysis of SL and the concept of the Metaverse to highlight how virtual worlds have evolved and how the Metaverse stands out from previous virtual worlds. This analysis contributes to a more balanced perspective of the Metaverse which is often suggested as a revolution although it is rather an evolution.

The rest of this paper is structured as follows: Section 2 embeds the paper in the discourse of virtual worlds and provides contextual knowledge on the development and status quo of both SL and the Metaverse. In Section 3 we present our methodological approach. Section 4 introduces a framework covering four key attributes of virtual worlds: immersiveness, social networking, persistence, and interoperability. In Section 5 we base our comparison of SL and the Metaverse on these attributes and explain where the Metaverse seems to stand out (and where not). In Section 6 we discuss the findings of our comparative analysis which lead us to conceptualize the Metaverse as an incremental innovation of multiple technologies maturing simultaneously to create an immersive and engaging user experience. We also discuss immersiveness as the potentially most important attribute contributing to the potential future success of the Metaverse. Finally, we conclude the paper in Section 7.

## 2. Background

### 2.1. Virtual worlds

The term "virtual worlds" has undergone significant evolution from the 2000 s till now (Downey, 2014; Girvan, 2018; González, Santos, Vargas, Martín-Gutiérrez, & Orihuela, 2013). Castronova (2001)'s work is one of the earliest comprehensive accounts of virtual worlds. He provides an in-depth analysis of "Norrath" and delves into its attributes. He talks about the level of interactivity amongst its players through avatars, its persistence and the immersive imagery in the game. Similarly, Jakobsson (2002) described virtual worlds as shared environments that are rendered on personal computers, and both the environments and users' identities were more persistent.

A well-established conceptualization of the virtual world from this era is from M. W. Bell (2008), who defined it as a synchronous, persistent network of people, represented as avatars, facilitated by networked computers. These virtual worlds were often characterized by their focus on social interaction and community building and were primarily used for entertainment and leisure activities.

As the technology and capabilities of virtual worlds advanced, the term began encompassing a broader range of applications. For example, in the mid-2000 s, virtual worlds became increasingly popular as educational and training tools (Boulos, Hetherington, & Wheeler, 2007; Salmon, 2009; Warburton, 2009). This was driven by the potential for immersive simulations to provide a realistic and engaging environment for learning, as well as the ability to facilitate collaboration and social learning among users (Nunes et al., 2017).

In recent years, the evolution of virtual worlds has continued, with new technologies and applications emerging. Virtual and augmented reality have become more mainstream, enabling users to interact with virtual worlds in more immersive and realistic ways (Di Natale, Repetto, Riva, & Villani, 2020). This has opened up new possibilities for various applications, from gaming and entertainment to education, healthcare, and beyond (Díaz, Saldaña, & Avila, 2020; Nunes et al., 2017).

While important attributes of virtual worlds - social interaction, immersion, persistence, interoperability - have remained constant, the emphasis and context of these attributes have shifted over time. In the early 2000 s, virtual worlds were primarily designed as social platforms and for gaming/entertainment. However, as technology developed further, virtual worlds were adopted for educational and training purposes, emphasizing the importance of immersion and realistic simulations. More recently, virtual worlds have evolved with new technologies, such as virtual and augmented reality, which have enabled even more sophisticated immersive experiences and new applications, such as virtual tourism and art, then were available in Second Life for example.

### 2.2. Second Life

Second Life (SL) was launched by Linden Lab in 2003, offering users a virtual world where they could create and personalize their online identities as avatars (Boellstorff, 2015; Enright, 2007). These avatars could interact with each other and the virtual environment, allowing users to express themselves in unique and creative ways (Fetscherin & Lattemann, 2007; Hemp, 2006). SL offered users the possibility of constructing an alternative identity that could either be a replication of their real-life self, an enhanced version with improvements along specific attributes, or a completely different self (Leonardi, 2020; McLeod & Leshed, 2011; McLeod, Liu, & Axline, 2014). Compared to other virtual worlds, users in SL faced no restrictions regarding the type of self-presentation that could be created, resulting in the situation where avatars could appear in any possible form and surround themselves with any objects of their liking (Leshed & McLeod, 2012). This freedom of expression and creativity was one of the key features that differentiated SL from other virtual worlds (Kafai, Fields, & Cook, 2010; Schechtman, 2012).

Moreover, users could also purchase virtual land and create private virtual spaces, such as houses, or personalize their avatars with virtual objects, such as clothes (Bloomfield & Cho, 2011; De Mesa, 2009). Additionally, SL provided a platform for users to buy and sell virtual and real products using Linden Dollars (L$), a virtual currency operated within the virtual world (Hendaoui, Limayem, & Thompson, 2008). The interesting aspect of L$ was that it had a real exchange rate with US dollars, making it possible for users to convert their virtual earnings into real currency. These features made SL a unique virtual world that offered a wide range of opportunities for users to express themselves and participate in virtual commerce.

However, despite its adoption in various industries, including healthcare training, finance, and education (Boulos et al., 2007; Salmon, 2009; Warburton, 2009), SL struggled to gain widespread popularity and experienced a decline in user numbers (with a short resurgence during the COVID19 pandemic). While it may not have achieved widespread popularity, SL still generates substantial revenue for Linden Labs and maintains a significant monthly user base. However, it faced and still faces challenges, such as a steep learning curve, clunky and laggy interface, persistent platform malfunctions and copyright violations, which have impeded its progress as a mass-market product (Jamison, 2017). While an upgrade to newer technologies like blockchain and NFTs may address some of these problems, this may be not easy given the legacy infrastructure of the platform (Schneider, 2020).

### 2.3. The Metaverse

The term "Metaverse" has many contradictory definitions but generally refers to a virtual world beyond the physical world. It was coined by science fiction writer Neal Stephenson in his novel "Snow-Crash," which describes a vast and densely populated virtual world that people can access through avatars (Joshua, 2017). One of the more popular definitions states that The Metaverse is characterized by the persistence of identity and objects, a shared environment, avatars, synchronization, three-dimensional space, interoperability, and an immersive and social user experience (Ball, 2022).

The concept of the Metaverse is considered a more advanced form of virtual experience, offering users a fully-immersive environment that integrates physical and virtual spaces (Ball, 2022). This is facilitated by using cutting-edge technologies such as virtual reality (VR), augmented reality (AR), artificial intelligence (AI), haptic feedback, blockchain, non-fungible tokens (NFTs), game engines, and the principles of Web 3.0, to name a few, which enable a more seamless, believable, and multi-sensory experience (Han, Bergs, & Moorhouse, 2022). The integration of these technologies is said to create a metaverse that is more economically and socially dynamic, allowing users to own and trade virtual assets, participate in decentralized communities, and engage in a new level of sophistication when it comes to virtual commerce (Crowell, 2022; Ryskeldiev, Ochiai, Cohen, & Herder, 2018; Valaskova, Vochozka, & Lăzăroiu, 2022). The use of game engines, such as Unity and Unreal Engine, may enable the creation of more erudite virtual environments and experiences, while Web 3.0 principles, such as decentralization and interoperability, can create a more open and accessible metaverse for users (Ryskeldiev et al., 2018). Additionally, using VR, AR, AI, and haptic feedback can create a fully immersive experience that blurs the line between the physical and virtual worlds, offering a new level of engagement and interaction (Valaskova et al., 2022).

## 3. Systematic literature review

Virtual worlds have become an increasingly popular platform for various applications, including education, training, and entertainment. As the number of users in virtual worlds continues to grow, there is a pressing need to understand the fundamental features and attributes that define these immersive environments. However, due to the diverse range of virtual worlds and the plethora of potential features that can be included, identifying the most salient and significant attributes can be challenging (M. W. Bell, 2008; Girvan, 2018). To address this challenge, a systematic literature review (SLR) is an appropriate methodology that can systematically and rigorously analyze a large body of existing research to identify virtual worlds' most prominent features and attributes (Okoli, 2015). This SLR aims to provide a comprehensive overview of the most prominent attributes of virtual worlds that are widely discussed and empirically supported in literature.

### *3.1. Search and extraction*

We considered IEEE, ACM Digital Library and Scopus search engine databases to execute the search string. The databases were chosen because they are of high quality and are peer-reviewed. Then, the authors discussed the keywords and the synonyms and reviewed the method to reduce the likelihood of biases or unintentional errors. Finally, we constructed a search string and used logical operators to complement the execution results. Each database has a different syntax, so it was necessary to create a specific query. Below is an example of one of those queries. TITLE (("Virtual Worlds") AND ("definition" OR "attribute" OR "feature")). One author conducted the search to include articles from January 2000 to January 2023.

We set specific practical screening criteria to confirm only quality publications included in the review at this stage. Therefore, invalid conference articles, reports, erratum, working papers, editorial notes, commentaries, and duplicated ones were excluded to emphasize journal publications and conference proceedings. The documents were carefully chosen based on their title and abstract. To provide a comprehensive review of the current literature, we did not exclude any studies based on their study designs; as a result, we included both qualitative and quantitative methodologies. The article did need to specifically discuss virtual worlds and not virtual work environments, for example.

The search also excluded reviews that were not written in English. The authors reviewed the search results independently and screened publications from databases and reference lists, following predefined steps. Initially, articles were screened based on their title and abstract. Then, to determine eligibility for inclusion in the review, we retrieved articles with potential relevance for full-text review. Finally, discussions were held to resolve disagreements.

We uploaded the selected articles to Excel to extract the data. In each article, each variable of interest was coded individually. For example, Okoli (2015) states that data extraction is essential and relevant to the research topic. The selected reviews were evaluated and coded to extract information such as study focus, year, definition, and attributes of virtual worlds. Coded data were clarified as needed until the study team reached a consensus.

### *3.2. Search outcomes*

First, we found 920 articles from the three databases based on search keywords, and after extracting only those from journals and conference papers, we were left with 563 articles. Second, we removed the duplicated studies, and researchers reviewed each journal, and conference proceedings that were recognized as potentially relevant and eliminated the others, therefore, 442 articles were removed. At the end of the next phase, the papers that did not fit the criteria were extracted, and 68 articles were obtained. Based on the PRISMA statement, Fig. 1 describes the methodology used to conduct this systematic review. PRISMA stands for "preferred reporting items for systematic literature review and meta-analysis" (Moher et al., 2009).

In order to identify the high-level attributes of virtual worlds, we analyzed 68 studies. We then elaborate on four attributes of virtual worlds which we have identified through inductive logic of reflection and abstraction.

## 4. Framework: key attributes of virtual worlds

### *4.1. Theoretical foundations*

We develop the framework which builds the theoretical basis for our comparative analysis through inductive processes of reflection and abstraction as described in Gregor, Müller, and Seidel (2013) 's approach for developing design theories. Whereas reflection refers to learning from past experiences, abstraction describes deriving generic features from observed instances of artifacts (J. S. Lee, Pries-Heje, & Baskerville, 2011).

Our comparative analysis is not intended to develop a complete design theory (which could happen as a next step). It can instead be understood as nascent principles of the design of virtual worlds (Heinrich & Schwabe, 2014). In line with our study's objective, we focus on the design theory component "Purpose and Scope". Gregor et al. (2013) suggest identifying the artifact's high-level meta-requirements or goals, which can be assisted by reflecting on the original problem-solution space.

This study is based on a sociotechnical perspective (Sarker, Chatterjee, Xiao, & Elbanna, 2019) where we focus not only on the technical features or tools available but also on the user experience offered by virtual worlds. By reviewing the academic literature above, we have identified four essential attributes of virtual worlds that serve as a framework for comparing and discussing SL and the Metaverse. These are immersiveness, social networking, persistence, and interoperability.

### *4.2. Framework*

Our framework is explained in more detail in this section. Briefly, immersiveness refers to the degree to which users can experience virtual worlds as if they were real, enabled by human-like interactions, haptic feedback, and virtual narratives (Ricci, 2020; Verhulst, Normand, Lombart, & Moreau, 2017). Secondly, social networking is a significant focus of virtual worlds, which is intended to be a social environment where users can interact in real life (Cheng et al., 2022; Falchuk, Loeb, & Neff, 2018). Third, persistence refers to the fact that the virtual content and experiences in virtual worlds remain accessible even after a user has logged out, creating a sense of continuity and a more engaging experience (Treem & Leonardi, 2013). Finally, interoperability is a key aspect of virtual worlds, which is designed to be accessible through a range of devices and technologies, including VR, AR, and WebXR technologies and across multiple virtual worlds and platforms (Havele, Brutzman, Benman, & Polys, 2022). (Tables 1, 2, 3, 4, and 5).

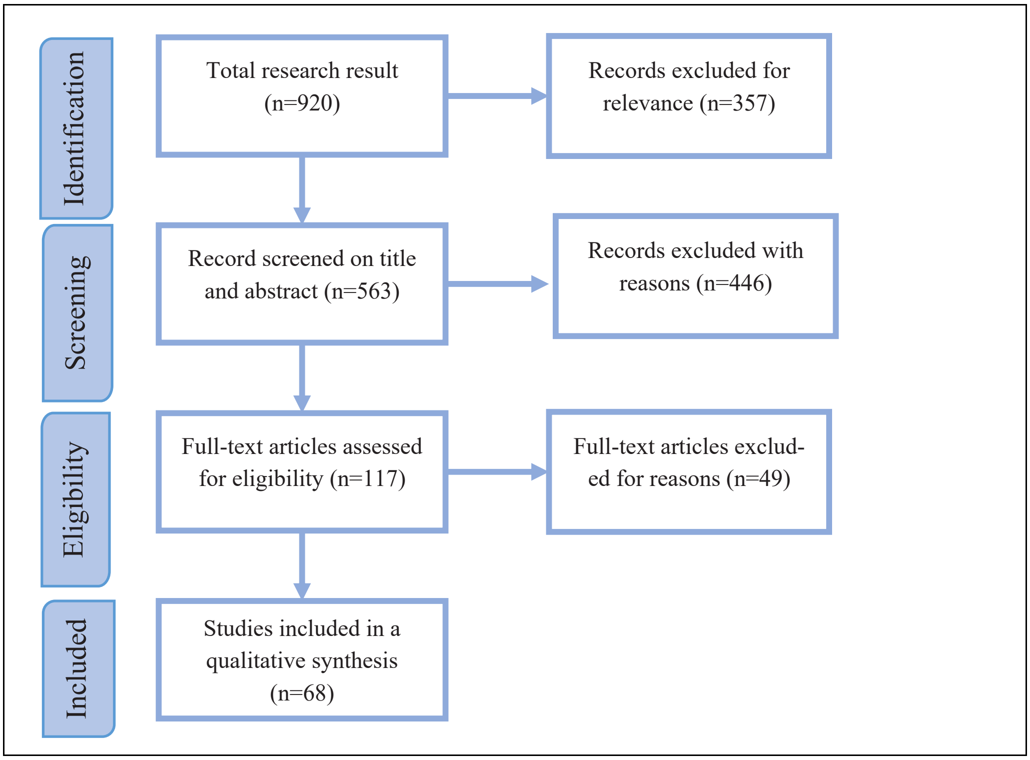


**Fig. 1.** PRISMA flow diagram describing the selection process. (Author's own creation).

**Table 1**
Defining the attribute Immersiveness.

| Attribute | Definition | Examples |
|---|---|---|
| Immersiveness | Perception of being physically present in a non-physical world | Losing themselves in the experience<br>Audio/visual and haptic feedback<br>Interacting with environment, socializing, and gameplay<br>Use of digital representations, such as avatars or virtual objects<br>Sensory environment through lighting and sound effects<br>VR enables use hands and body to perform virtual actions<br>AR integrates virtual objects with the physical world for user interaction |

**Table 2**
Defining the attribute Social Networking.

| Attribute | Definition | Examples |
|---|---|---|
| Social Networking | Intensified level of social engagement and community formation that is attainable for users within these digital spaces | Text-based and voice-based communication<br>Facial cues<br>Integration of social media platforms<br>Social spaces eg parties, concerts, lectures<br>Virtual marketplace/ virtual economy |

## 4.3. Immersiveness

Immersiveness refers to the extent to which users can "immerse" themselves in the virtual world, feel as though they are physically present within it (Han et al., 2022) and interact in such a manner that they

**Table 3**
Defining the attribute Persistence.

| Attribute | Definition | Examples |
|---|---|---|
| Persistence | Virtual creations, interactions, and relationships established within the virtual environment can be saved and maintained even after the user has logged out | The creation or adaptation of parts of the world persists once the user has logged off<br>Ability to build and leave behind a structure/community/office space for others to use to reuse |

**Table 4**
Defining the attribute Interoperability.

| Attribute | Definition | Examples |
|---|---|---|
| Interoperability | Refers to the ability of the virtual environment to be accessed and experienced across different virtual worlds and platforms | Creation of an avatar or digital object that can be carried from one platform/ world to another<br>One currency that is valid across all worlds and platforms |

lose track of time (Goel, Johnson, Junglas, & Ives, 2013). It has also been referred to the degree to which people perceive that they interact with their virtual environment rather than their physical surroundings (Guadagno et al., 2007). The use of immersive technology can enhance the immersion level, providing users with a more realistic and engaging experience, and making it easier for them to suspend their disbelief and fully immerse themselves in the virtual world (Di Natale et al., 2020; Slater, 2018).

An immersive virtual environment can provide a believable, realistic, and engaging experience for users (Dincelli & Yayla, 2022). This can be achieved through a combination of factors. The sensory feedback that users receive in response to their actions within the virtual world, whether through audio/visual or haptic means, plays a significant role

**Table 5**
Comparative Analysis of Second Life and the Metaverse.

| Attribute | Application in Second Life | Application in the Metaverse |
|---|---|---|
| Immersiveness | Immersive environment limited by LSL (Linden Scripting Language) which can cause lag. | Cutting-edge technologies like AR/VR, haptic feedback that create hyper-realistic worlds. |
| Social Networking | Most communicative affordances are technically restricted to a greater or lesser extent.<br>A steep learning curve means users take time to figure out how to navigate and communicate appropriately | Presence of high fidelity in terms of settings and avatars' gestural and mimic expressiveness<br>Social Presence enhanced by VR/AR |
| Persistence | The virtual world itself is persistent, but persistence for avatars only exists when they are in-world | Users' virtual creations and interactions are more likely to be saved and persist over time, allowing for a greater sense of continuity and permanence |
| Interoperability | Limited success when a world shared the SL protocol or was on a Linden Lab owned server | Interoperability thus far not possible across multiple platforms like Roblox, Sandbox for example<br>Limited success with blockchain technologies |

in creating the feeling of actively participating and causing changes within the environment (Gibbs, Gillies, & Pan, 2022). This "cause and effect" feedback creates a sense of spatial presence and encourages users to feel more invested in the virtual world (Baños et al., 2004; Elmezeny, Edenhofer, & Wimmer, 2018). Similarly, the sensory environment, through lighting, sound effects, and other cues, also plays a significant role in creating an immersive experience. Tamborini and Skalski (2012) uses the example "a player swerving onto a rumble strip in a driving game might be met with thumping sounds plus vibrations in the game controller that match feelings in real life".

Users must be able to engage with the virtual environment in various ways, such as exploring different areas, interacting with other users, and playing games to fully immerse themselves in the virtual world (Calleja, 2014). This creates a more personalized and engaging experience promoting a sense of presence and investment in the virtual world (Han et al., 2022; Violante, Vezzetti, & Piazzolla, 2019).

Using digital representations, such as avatars or virtual objects, further enhances the sense of immersion. For example, Waltemate, Gall, Roth, Botsch, and Latoschik (2018) found that personalized avatars significantly increase body ownership, presence, and dominance compared to their generic counterparts. Moreover, a study by Lau and Lee (2015) found that using avatars enabled a more immersive and, in turn, effective learning environment for students.

Virtual reality (VR) technology elevates the concept of immersion by enabling users to use their hands and body to perform virtual actions, which creates a more natural and realistic experience. Various professionals leverage virtual reality technologies for different purposes. Cultural Heritage sites, for instance, utilize virtual reality to situate their visitors within historical stories, creating a more immersive and engaging reading experience (Škola et al., 2020). Similarly, educators use virtual technologies for experiential learning, allowing students to participate in simulations and acquire practical skills in a risk-free environment. In psychiatry, virtual reality is used to mitigate the adverse effects of psychological traumas, providing patients with exposure therapy that can reduce anxiety and improve their overall mental health (Markowitz & Bailenson, 2019).

Augmented reality (AR) integrates virtual objects with the physical world, allowing user interaction and creating a sense of integration between the digital and physical worlds (Rauschnabel, Rossmann, &. Companies like IKEA, Nike, ASOS, and Amazon have incorporated augmented reality technology to enhance their products' immersive experience (McLean & Wilson, 2019). By combining real and virtual objects with stereo tracking technology, AR-based simulators can significantly improve immersion into the surgical environment and the accuracy of perception of operation regions (Si et al., 2019).

A highly immersive environment can create a sense of presence, or the illusion of being physically present in the virtual world, leading to increased user engagement and enjoyment (D. Shin, 2019; Violante et al., 2019). Additionally, the level of immersion in a virtual environment has been shown to positively impact various cognitive and affective responses, such as memory retention, perception, emotion, and motivation (Visch, Tan, & Molenaar, 2010; Winkler, Röthke, Siegfried, & Benlian, 2020).

### 4.4. Social networking

Social networking in virtual worlds refers to the intensified level of social engagement and community formation attainable for users within these digital spaces. Social presence is defined by Biocca and Harms (2002) as "the sensation of being with another person in a mediated environment," and by Blascovich et al. (2002) as "the extent to which an individual believes that they are interacting with other genuine human beings." Goel et al. (2013) similarly refer to this as the "perception of others as being in the same space," and they call this social perception. Essentially, these researchers suggest that environments facilitating immersion and incorporating social presence are more akin to face-to-face interactions and provide a novel avenue for exploring human interaction.

From a user experience perspective, this concept emphasizes the potential for virtual worlds to provide an immersive and highly social environment where individuals can connect and interact with others in meaningful ways (Cheng et al., 2022). Virtual events, multiplayer games, co-creation activities, and various forms of digital communication, such as text-based chat and voice chat, are just a few examples of the numerous social experiences available within virtual worlds (Buana, 2023). These multi-modal interaction features give users in the virtual gaming environment an opportunity to collaborate to solve quests, build and create structures and worlds, or compete against each other (Consalvo, 2017). This multi-modality in a 3D environment also contributes to engaging creative team collaboration and social interaction in a work environment (Alahuhta, Nordb, Sivunen, & Surakka, 2014). These activities offer users the opportunity to form relationships, participate in activities, and engage in collaborative projects with others worldwide, creating thriving virtual communities (Hennig-Thurau et al., 2022).

Compared to the physical world, virtual worlds provide users with increased flexibility and accessibility regarding social interaction. For instance, virtual social events and multiplayer games can occur 24/7, allowing users to connect and engage with others from different time zones (Kohler, Fueller, Matzler, Stieger, & Füller, 2011). Additionally, virtual worlds can provide social experiences that may not be possible in the physical world, such as avatar-based experiences and virtual reality games (Yeh, Chuan-Chuan Lin, & Lu, 2011). Social networking in virtual worlds significantly shifts how we approach social interaction and community formation.

In particular, Goel et al. (2013) assert that simply creating 3D virtual places, such as malls or offices, will not be enough to attract people to return to a virtual world, particularly if these spaces remain empty. Instead, virtual environments must prioritize social experiences that allow avatars to interact with each other in a meaningful way. For example, merely seeing others in the virtual space is insufficient; the quality of these interactions leads to the experience of flow and encourages individuals to revisit the environment (Goel et al., 2013). This underscores the significance of hosting in-world events that offer opportunities for avatars to engage with one another. By providing users with the tools and opportunities to connect and engage with others in meaningful ways, virtual worlds can change how we approach social experiences, creating new and innovative opportunities for connection and community building.

### 4.5. Persistence

Persistence refers to the idea that the virtual environment and its actions persist over time (Treem & Leonardi, 2013). Bell (2008) describes persistence as the idea that a virtual world cannot be paused" or does not shutdown when the user exits. The persistence of virtual worlds, as opposed to video games, allows for a change in how people interact with other participants and the environment (Ball, 2022). Instead of being the center of the world, participants become part of a dynamic community and evolving economy. In addition, this persistence creates a sense that the systems within the virtual world exist independently of the participant's presence (Bell, 2008).

Persistence in virtual worlds means that virtual creations, interactions, and relationships established within the virtual environment can be saved and maintained even after the user has logged out (Dwivedi et al., 2022). For example, suppose a user creates an object within the virtual world and leaves it in a specific location. In that case, other users can view that object at the same location, regardless of whether the first user is online. However, if a second user removes the object from that location, the first user will not be able to see it when they return to the location. (Girvan, 2018).

This creates a sense of continuity and a permanent digital presence within the virtual world. Persistence is a key aspect of any virtual world, as it allows users to build and maintain their virtual identity and experiences, creating a more immersive and engaging virtual environment (Weinberger, 2022). Additionally, persistence can facilitate the creation of long-lasting virtual communities and the continuation of virtual relationships and interactions over time (Nevelsteen, 2018). For example, in an education study by Girvan and Savage (2019), it was found that the quality of persistence was a significant educational affordance, as it enabled learners to observe the progress of "other installations" over time, even when the creators were not online. In addition, it facilitated a means for learners to receive feedback from peers and contribute to creating their artifact while their partner was offline.

Moreover, certain learning activities may prove difficult to execute in a physical environment due to high costs, lack of infrastructure availability, or potential risks. However, virtual worlds provide a solution by allowing for the simulation and execution of various activities. For example, The Polytechnic University at Madrid has designed multiple virtual worlds featuring practice laboratories. These virtual environments enable students to conduct laboratory experiments in various fields, including electronic systems, materials, biotechnology, chemistry, and more, without requiring them to attend a physical laboratory (González et al., 2013). Instead, instructors can create a laboratory that can be utilized multiple times by several students and can be reset after a student has completed their work much like a real lab.

### 4.6. Interoperability

Interoperability refers to the ability of the virtual environment to be accessed and experienced across different virtual worlds and platforms (Havele et al., 2022). The goal of interoperability is for users to have greater accessibility and flexibility, allowing them to continue their virtual experiences seamlessly (Verhulst et al., 2017). By offering interoperability, virtual worlds aim to provide users with a more immersive and engaging virtual experience that is accessible across boundaries between virtual worlds (Gadekallu et al., 2022).

Interoperability is crucial in advancing the virtual world experience, as it enables users to access and participate in virtual activities, content, and communities across different virtual environments (Jovanova & Preteux, 2009). This broadens the scope of the virtual experience and fosters greater collaboration and interaction among users across virtual boundaries (Lorenzo, 2011). Additionally, interoperability can support the development of more sophisticated virtual environments and applications, enabling greater interoperability between virtual objects, applications, and services (J. Bell, Dinova, & Levine, 2009).

Although numerous platforms, such as Roblox, The Sandbox, and Decentraland, have established thriving virtual worlds, they are limited by their lack of interoperability. This hinders the creation of a seamless, integrated Metaverse experience, as envisioned by Neal Stephenson in his seminal novel (Joshua, 2017). Thus while, the current state of virtual worlds has been praised for its innovative approach to gaming, social interaction, and content creation (Adams, 2022; Ahn, Kim, & Kim, 2022; Branca, Resciniti, & Loureiro, 2022; Han et al., 2022). The lack of interoperability is a crucial aspect that must be addressed to achieve the grand vision of virtual worlds like the Metaverse (Havele et al., 2022; Lorenzo, 2011). Each platform operates in isolation, creating fragmented virtual experiences that are not interconnected. This is in stark contrast to the Metaverse concept, which imagines a unified virtual environment where users can seamlessly move from one world to another, interacting with other users, objects, and information (Xu et al., 2022).

## 5. Comparative analysis

We will now compare SL and the Metaverse based on the four dimensions of our framework of key characteristics of virtual worlds. As introduced in the previous section we apply inductive reasoning by reflecting and abstracting on the key characteristics of virtual worlds.

### 5.1. Immersiveness

In terms of immersiveness, SL offered a basic level of immersion compared to the Metaverse. SL provides interaction with avatars and digital objects. Still, it does not have the same level of physical immersion as the Metaverse, which allows for real-life movements and interactions within the virtual environment. This is perhaps because SL depends on Linden Scripting Language to create the environment, which is somewhat more limited when compared to the advanced technologies that are now available to develop these hyper-realistic worlds (Hai-Jew, 2023).

In the context of the Metaverse, cutting edge technologies such as AI, blockchain, 5 G/6 G, real-world graphics rendering, and digital twins, among others (Balica, Majerová, & Cuțitoi, 2022; Mozumder, Sheeraz, Athar, Aich, & Kim, 2022; Valaskova et al., 2022; Xu et al., 2022) enable the creation of a truly immersive virtual environment that offers a seamless connection between the virtual and physical world. They allow for hyper-realistic graphics, intuitive haptic feedback, and AI-driven interactions that enhance the overall user experience in the Metaverse (L.-H. Lee et al., 2021; Xu et al., 2022). These technologies work together to create a virtual world that is not only believable, but also highly engaging and offers a level of immersion that was previously unattainable (Havele et al., 2022).

### 5.2. Social networking

Both the Metaverse and SL, offering the potential for highly social virtual experiences can be compared in terms of their socialness, with both platforms. Both environments allow users to interact with others in various ways, such as through text-based chat, voice chat, and avatar-based social interactions (Faiola & Smyslova, 2009; Kaplan & Haenlein, 2009). SL provides users with basic tools for communication and interaction, allowing for social networking experiences (Dionisio, III, & Gilbert, 2013). However, the Metaverse takes this to a new level, integrating advanced communication technologies that enable more natural and immersive user interactions. The Metaverse also expands the scope of social interactions, allowing for a broader range of virtual experiences, from virtual concerts to team-based gaming (Meluso, Johnson, & Bagrow, 2020) and new opportunities for connection and community building (Bousba & Arya, 2022).

Previous studies found that constructing identities in SL could be disorienting and complex, as identities are fluid and malleable

(Warburton, 2009). This created challenges in building social relationships as reputation management and accountability became concerns. Additionally, SL's unique cultural norms and codes could be challenging to navigate and contribute to a sense of isolation. Finally, the lack of in-world persistence deters casual use and can create a destabilizing environment with few boundaries or limits on behavior (Meadows, 2007). Because the concept of the Metaverse is not yet mature enough, we do not yet know if users will run into similar concerns as users of early SL versions.

### 5.3. Persistence

The Metaverse and SL can be compared in terms of persistence, with both platforms offering a persistent virtual environment where users can create, interact, and persistently maintain their digital presence (González et al., 2013). In SL, user-generated content and virtual environments could persist over time, providing users a sense of ownership and investment. This set SL apart from its contemporaries; persistent spaces and customizable avatars created an environment where time and space held significance. Individuals could inhabit and transform this virtual world by actively participating in it. However, the persistence of virtual environments in the Metaverse has been enhanced, offering a more stable and enduring virtual environment that enables users to build and maintain their virtual presence (Dionisio, W. G. B., & Gilbert, 2013). In the Metaverse, users' virtual creations and interactions are more likely to be saved and persist over time, allowing for a greater sense of continuity and permanence. Additionally, the Metaverse may allow for greater cross-platform persistence, enabling users to continue their virtual experiences across multiple devices and platforms (Grieves & Vickers, 2017; Weinberger, 2022).

### 5.4. Interoperability

Interoperability, was a matter of great debate when SL and its derivatives like OpenSim were introduced (J. Bell et al., 2009; Lorenzo, 2011). Similarly, the Metaverse has also made interoperability a key focus. Eventually enabling users to access and engage with virtual environments across a wide range of platforms and devices, allowing users to continue their virtual experience and keep their digital identity consistent across many different virtual worlds (Xu et al., 2022). While there were some primitive moves towards achieving interoperability between different virtual worlds in the 2000 s, there were similar roadblocks to those that exist today.

For example, the underlying architectures and protocols for many virtual worlds can be different; therefore, agreeing on a standard way of developing a virtual world was and is contentious (Thompson, 2010). There was limited success when a world shared the SL protocol or was on a Linden Lab owned server (Thompson, 2010, 2011).

While interoperability between virtual worlds in the Metaverse has yet to occur, to our knowledge (L.-H. Lee et al., 2021), there do exist technologies now that could, in theory, help with this problem. Blockchain technology has emerged as perhaps the most plausible solution to the interoperability of virtual worlds (Belchior, Vasconcelos, Guerreiro, & Correia, 2021; Mozumder et al., 2022; Ryskeldiev et al., 2018). More specifically, cross-chain protocols would enable the exchange of possessions like avatars, NFT's, and payment between virtual worlds (Belchior et al., 2021; Madine et al., 2021). However, this technology has its challenges in its current format. Gadekallu et al. (2022) share two of these challenges: the presence of multiple public blockchains operating in different virtual worlds, with no shared language between them, and the transaction architecture and consensus processes used in these virtual worlds differ significantly, which further hampers interoperability.

## 6. Discussion

### 6.1. Implications for research

Based on our comparative analysis, the concept of the Metaverse is not fundamentally different from previous implementations of virtual worlds like SL. It can instead be seen as an incremental improvement but in such a way that it breaks through user resistance. Most importantly, the improvements in the dimensions of immersiveness may mark an essential cornerstone of the future success of the Metaverse. While some of the elements of the Metaverse existed in SL, the advanced technology used in creating the Metaverse provides a much more immersive and engaging virtual environment, enhancing social interaction and creating a seamless connection between the virtual and physical worlds. The Metaverse offers researchers an opportunity to explore the evolution of virtual worlds, the factors that contribute to the success or failure of virtual platforms, and identify best practices and potential pitfalls for the development and implementation of virtual environments.

### 6.2. Implications for practice

All the elements of persistence, social networking, interoperability, avatars, and immersive-ness were present to some extent in Second Life (SL). However, SL represented a more limited and primitive form of virtual reality compared to the advanced and immersive virtual environments offered by the Metaverse. The key difference between SL and the Metaverse lies in the level of immersiveness and the underlying technological advancement (Rymaszewski et al., 2007). The Metaverse is designed to offer a truly immersive experience that enhances social interaction and creates a seamless connection between the virtual and physical worlds (Kohler et al., 2011; D. H. Shin, 2009). This is made possible by cutting-edge technologies such as AI, blockchain, and 5 G/6 G, which provide a more believable and engaging virtual environment. These advanced technologies enable the Metaverse to create realistic and interactive experiences that blur the boundaries between the virtual and physical realms (Kar & Varsha, 2023).

On the other hand, SL was built on existing web technologies in the early 2000 s, utilizing HTML, JavaScript, and PHP, and relied on basic computer graphics to create its virtual environment (Kaplan & Haenlein, 2009). The lack of access to advanced technologies such as AI, blockchain, 5 G/6 G, or real-world graphics rendering limited the immersive potential of SL (Warburton, 2009). SL was constrained by the hardware available at the time, including computer screens that did not offer the same level of resolution and graphical capabilities as today's XR (extended reality) devices. Consequently, SL fell short in providing a truly immersive and seamless interaction between the virtual and physical worlds (Love, Ross, & Wilhelm, 2009).

The implications for practice and policy are significant. The development of the Metaverse necessitates investment in cutting-edge technologies and infrastructure, including AI, blockchain, and advanced networking capabilities such as 5 G and future generations like 6 G. These advancements will not only enhance the user experience but also enable a wide range of applications and services within the Metaverse, spanning various sectors such as education, healthcare, entertainment, and business (Kohler et al., 2011). From a practical standpoint, businesses and organizations can leverage the immersive nature of the Metaverse to create novel and engaging experiences for their customers. For example, virtual showrooms and immersive shopping experiences can be developed, allowing customers to explore and interact with products in a virtual environment (Ball, 2022). Additionally, the Metaverse can provide new avenues for collaboration, enabling geographically dispersed teams to work together in virtual spaces (Grzegorczyk, Mariniello, Nurski, & Schraepen, 2021). These collaborative environments can foster creativity, productivity, and innovation.

On the policy front, governments and regulatory bodies need to address privacy, security, and ethical concerns associated with the

Metaverse. As users spend more time in virtual environments and generate vast amounts of personal data, safeguarding user privacy becomes crucial. Clear guidelines and regulations must be put in place to protect user data and ensure responsible use of emerging technologies (Rymaszewski et al., 2007).

Therefore, while SL laid the foundation for the development of the Metaverse, it was a more limited and primitive virtual reality platform compared to the immersive and advanced virtual environments offered by the Metaverse. The underlying technological advancements in AI, blockchain, and 5 G/6 G have enabled the Metaverse to create a truly immersive experience that seamlessly integrates the virtual and physical worlds. Embracing these advancements and addressing associated challenges will have wide-ranging implications for various sectors and require careful consideration of policies and regulations to ensure the responsible and ethical development of the Metaverse.

### 6.3. Future research directions

While the implementation of the Metaverse is still in its early stages, its potential and limitations are yet to be fully realized (Dionisio et al., 2013). The integration of new technologies, such as AR/VR and blockchain, offers exciting possibilities for the future development of the Metaverse, but also raises questions about the ethical implications of a highly-immersive, persistent and interoperable virtual environment (Fernandez & Hui, 2022). As the concept of the Metaverse continues to evolve, it will be important for researchers to consider the social, technological, and ethical implications of this new and rapidly evolving technology (Kshetri & Dwivedi, 2023). For example, privacy, security, and governance issues will need to be addressed, as will concerns around digital divide and accessibility (Falchuk et al., 2018; Fernandez & Hui, 2022). Furthermore, there is a risk of creating a fragmented and exclusive metaverse, where certain segments of society are excluded or marginalized (Bibri, 2022; Park, Ahn, & Lee, 2023).

To avoid repeating past mistakes and ensure that the Metaverse positively impacts society, it is important to consider the lessons learned from previous virtual world platforms, such as SL. For example, SL struggled to maintain user engagement and generate revenue, ultimately failing to live up to the hype surrounding it (Henninger et al., 2019). This highlights the importance of setting realistic expectations and avoiding overpromising and underdelivering. Additionally, it is important to approach the development and implementation of the Metaverse in a transparent, equitable, and inclusive manner, to ensure that the benefits of the technology are widely shared and accessible to all segments of society.

The potential impact of the Metaverse on society is significant and far-reaching. Unlike previous virtual worlds, such as SL, the Metaverse promises to offer a much more immersive and social experience. This new level of immersion and social interaction has the potential to transform the way people communicate and interact with each other, as well as the way they experience and participate in commerce, entertainment, education, and other aspects of society (Buana, 2023; Cheng et al., 2022; Hennig-Thurau et al., 2022).

While the potential impact of the Metaverse on society is significant, its limitations and potential negative consequences must also be considered. The ethical implications of a highly immersive, persistent, and interoperable virtual environment require further exploration. Researchers must continue to investigate the social, technological, and ethical implications of the Metaverse and consider the lessons learned from previous virtual world platforms. The integration of new technologies, such as AR/VR and blockchain, offers exciting possibilities for the future development of the Metaverse, but also raises questions about the ethical implications of a highly immersive, persistent, and interoperable virtual environment.

Based on our study, we suggest the following research avenues in form of research questions:

*What measures can be taken to address privacy, security, and governance issues in the development and implementation of the Metaverse?*

To address privacy, security, and governance issues in the development and implementation of the Metaverse, comprehensive measures are required. This includes robust data protection mechanisms, user consent frameworks, and transparency in data collection and usage (Fernandez & Hui, 2022). Collaboration between governments, regulatory bodies, industry stakeholders, and user communities is essential to develop and enforce these measures effectively (Falchuk et al., 2018).

*How can the lessons learned from previous virtual world platforms, such as Second Life, inform the development and implementation of the Metaverse to avoid similar pitfalls?*

Lessons learned from previous virtual world platforms, such as Second Life, can inform the development and implementation of the Metaverse to avoid similar pitfalls. These lessons include the need for improved user experience, enhanced graphical capabilities, seamless integration of the virtual and physical worlds, and the establishment of clear guidelines for content creation and community management (Kanamgotov et al., 2012; Schechtman, 2012).

*In what ways can transparency, equity, and inclusiveness be incorporated into the development of the Metaverse to ensure widespread access to its benefits?*

Incorporating transparency, equity, and inclusiveness into the development of the Metaverse is crucial to ensure widespread access to its benefits. This can be achieved through transparent decision-making processes, open standards, addressing affordability and accessibility barriers, compatibility with a range of devices, and actively engaging with marginalized communities during the design and implementation phases (L. B. L. Rosenberg, 2022; L.B. Rosenberg, 2022; Zallio & Clarkson, 2022).

*How might the increased immersion and social interaction offered by the Metaverse transform various aspects of society, such as commerce, entertainment, and education?*

The increased immersion and social interaction offered by the Metaverse have the potential to transform various aspects of society, including commerce, entertainment, and education. Virtual marketplaces can revolutionize shopping experiences, immersive storytelling can enhance entertainment, and immersive learning environments can facilitate collaborative education experiences (Bibri, 2022; Cheng et al., 2022; Hennig-Thurau et al., 2022).

*What potential negative consequences and ethical implications may arise from a highly immersive, persistent, and interoperable virtual environment like the Metaverse, and how can they be addressed?*

While the Metaverse offers significant potential, there are potential negative consequences and ethical implications that must be addressed. These include issues of addiction, loss of privacy, virtual identity theft, cyberbullying, and the potential reinforcement of societal biases within virtual environments (Dolata & Schwabe, 2023; Kshetri & Dwivedi, 2023). Ethical guidelines, responsible platform governance, user protection, and content moderation are necessary to mitigate these concerns (Fernandez & Hui, 2022).

*How can the integration of new technologies, such as AR/VR and blockchain, in the Metaverse be balanced with ethical considerations and potential risks?*

The integration of new technologies, such as AR/VR and blockchain, in the Metaverse should be balanced with ethical considerations and the assessment of potential risks. Ethical frameworks should govern the use of personal data, immersive technologies, and the implementation of blockchain-based systems (Fernandez & Hui, 2022; L. Rosenberg, 2022; L.B. Rosenberg, 2022).

*What are the long-term psychological and social implications of increased immersion in the Metaverse on individual users and society as a whole?*

The long-term psychological and social implications of increased immersion in the Metaverse on individual users and society as a whole should be studied (Mogaji et al., 2023). Research is needed to understand the effects on mental health, social relationships, and human behavior (Buana, 2023). Educational initiatives, digital well-being programs, and user awareness campaigns can help users navigate the challenges associated with increased immersion (Dincelli & Yayla, 2022; Mogaji et al., 2023).

*How can researchers and developers collaborate to create a comprehensive ethical framework for the development and implementation of the Metaverse?*

Researchers and developers should collaborate to create a comprehensive ethical framework for the development and implementation of the Metaverse. This collaboration should include multidisciplinary research, consultation with diverse stakeholders, and addressing aspects such as privacy, security, governance, inclusiveness, and transparency (Maier, Laumer, & Weitzel, 2022; Maier, Thatcher, Grover, & Dwivedi, 2023).

## 7. Conclusion

The Metaverse is an evolving concept that incorporates the latest advancements in fields such as virtual and augmented reality, haptic feedback, artificial intelligence, and blockchain technology (Dincelli & Yayla, 2022; Havele et al., 2022; Mozumder et al., 2022). The development of the Metaverse builds upon the foundation laid by earlier virtual world platforms, such as SL. Building on a theory-driven framework, we showed in this paper how advancements in technology, including VR and AR, will enable a more immersive and interactive experience, allowing users to engage with virtual environments more comprehensively and interact with other users more intuitively and naturally. As a result, the Metaverse, which has not yet come into existence, has the potential to transform how we communicate, work, and socialize, and its impact on society will likely be far-reaching and profound.

## CRediT authorship contribution statement

**Shahper:** Data curation, Formal analysis, Methodology, Investigation, Writing – original draft, Writing – review & editing. **Alexander:** Conceptualization, Methodology, Investigation, Writing – original draft, Writing – review & editing. We would like to express our collective commitment to transparency and acknowledge the unique and shared contributions of Shahper and Alex to the research and preparation of this paper.

## Declaration of Competing Interest

The authors declare no conflict of interest. The research was not funded by any external sources.

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